EDV21C 系列产品使用说明书

EDV21C Series User Manual

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产品介绍 / Product Introduction

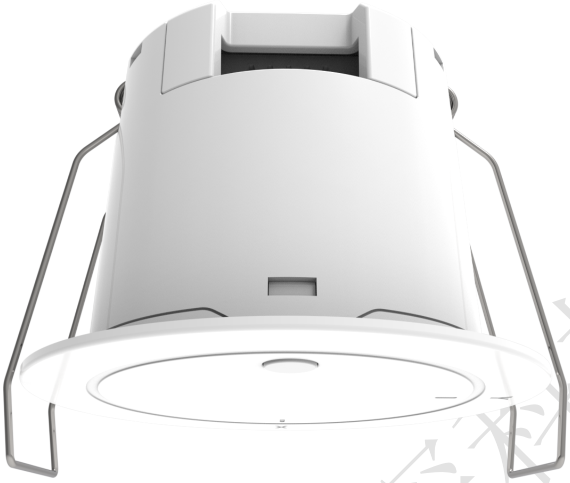

图 1.1: 正面视图
Fig1.1: Front View

图 1.2: 背面视图
Fig1.2: Reverse View

1.1 产品概述 / Overview

EDV21C 系列产品，工作频段为 59-64GHz，主要面向区域人数统计与人员轨迹监测应用，单台设备能够实现 3 米范围内区域局部分区，对区域内人员存在状态、移动轨迹及在场人数进行同步探测。产品融合纯毫米波 FMCW 探测技术，依托雷达捕捉人体微动、行走动作，可同时追踪最多 20 组人员坐标轨迹；并通过 RS485、KNX、Wi-Fi-MQTT 等多通讯渠道实时上报全部检测数据。设备不受光照、温湿度变化干扰，无图像采集，从根源保护用户隐私，搭配三维分区屏蔽功能可过滤家具、墙体等静态干扰，适用于商业安防、酒店商用、办公空间等场景的人员监测。

The EDV21C series products operate in the 59 - 64 GHz frequency band, primarily designed for regional crowd counting and human trajectory monitoring applications. A single device supports local zone segmentation within a 3-meter range, enabling simultaneous detection of human presence status, movement trajectories and the number of occupants in the monitored area. Adopting pure millimeter-wave FMCW detection technology, the products capture tiny human movements and walking motions via radar, and can track up to 20 sets of human coordinate trajectories concurrently. All detected data is reported in real time through multiple communication interfaces including RS485, KNX and Wi-Fi-MQTT.Free from interference by light, temperature and humidity fluctuations, the devices collect no images to fundamentally protect user privacy. Equipped with a 3D zone shielding function to filter static interferences such as furniture and walls, they are suitable for human monitoring scenarios including commercial security, hotel premises and office spaces.

1.2 产品特点 / Features

- 灵活配置：多配置方式、丰富配置内容、配置便捷性；搭配蓝牙小程序，可在线完成三维分区、灵敏度、延时、网络通讯等全量参数调试与自定义设置，操作简单直观。
  Flexible Configuration: Multiple configuration modes, abundant configuration items, and convenient configuration operations. Paired with a Bluetooth mini-program, users can complete full-parameter debugging and custom settings online, including 3D zoning, sensitivity, latency, network communication and more, with simple and intuitive operations.

- 抗干扰能力强：搭载 60GHz 高抗干扰天线，运行稳定，有效抵御光照、温湿度、绿植窗帘无热源移动、2.4G 无线设备等各类环境干扰。
  Strong anti-interference capability: Equipped with a 60 GHz high anti-interference antenna, it operates stably and effectively resists various environmental interferences such as light, temperature and humidity, heat-free movement of green plants and curtains, and 2.4G wireless devices.

- 供电方式灵活：交流、直流多种供电方式可选，均支持宽电压供电
  Wide-voltage power supply: -W supports AC 90-270V, while -4/-K support DC 12-30V, suitable for various power supply environments.

- 易安装部署：采嵌入式独立安装，标准开孔 Φ55mm，施工便捷。
  Easy installation and deployment: It adopts embedded independent installation, with standard opening diameter of Φ55mm, making construction very convenient.

- 通讯拓展丰富：分型号搭载 Wi-Fi、RS485、KNX 多类通讯协议，可无缝对接楼宇自控、智能家居、物联网云平台等各类智能控制系统。
  Rich communication expansion: Different models are equipped with various communication protocols including Wi-Fi, RS485 and KNX, enabling seamless connection to diverse intelligent control systems such as building automation, smart home and IoT cloud platforms.

1.3 应用场景 / Applications

- 智慧办公：开放式工位分区人数统计，人在灯亮、人走灯灭节能管控。
  Smart Office: Statistics on the number of people in open workstation zones, energy-saving control with lights on when occupied and lights off when vacant.

- 智慧酒店：客房人员数量识别、有人无人判定，联动客房用电节能管控。
  Smart Hotel: Automatically identifies the number of staff in the rooms, determines whether there are people or not, and links with the energy-saving management of room electricity usage.

- 智能家居：全屋房间存在感知、分区人员监测，联动照明、空调自动运行。
  Smart Home: All the rooms in the house have sensing capabilities, personnel monitoring by zones, and the lighting and air conditioning systems operate automatically in coordination.

- 医疗康养：实时监测老人与康复患者活动轨迹，辅助康复状态监测。
  Medical care and health care: Real-time monitoring of the activity trajectories of the elderly and rehabilitation patients, assisting in the monitoring of their rehabilitation status.

1.4 外形结构 / Appearance Structure

1.4.1 外形尺寸 / Dimension

全系列通用安装开孔：标准 Φ55mm，产品整体尺寸公差 ±0.2mm。
Universal mounting hole for the whole series: standard Φ55mm; overall dimension tolerance ±0.2mm.

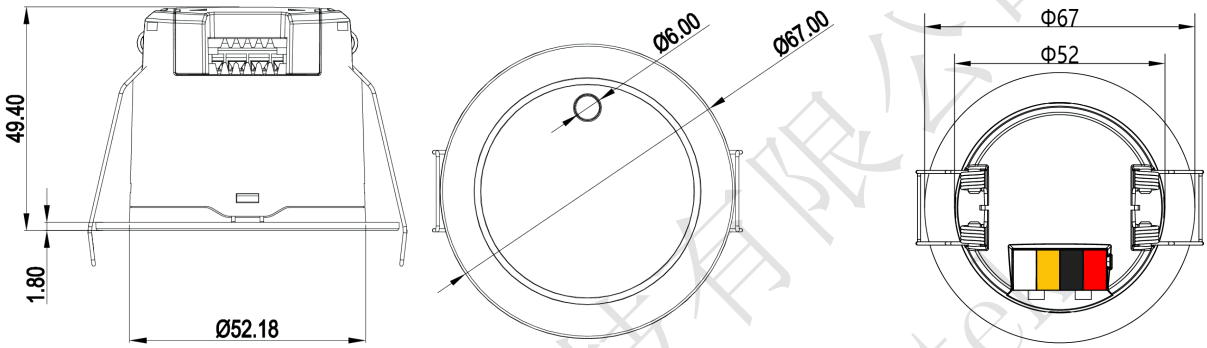

图 1.3: 单位（mm）
Fig1.3: Unit (mm)

1.4.2 接口说明 / Electrical Interface

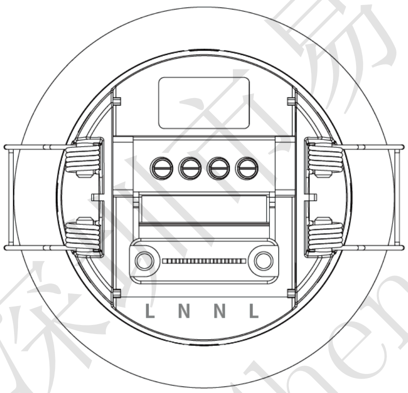

图 1.4: EDV21C-W 接口分布图
Fig1.4: EDV21C-W Interface Distribution Diagram

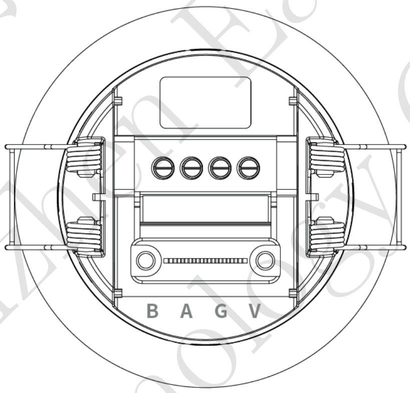

图 1.5: EDV21C-4 接口分布图
Fig1.5: EDV21C-4 Interface Distribution Diagram

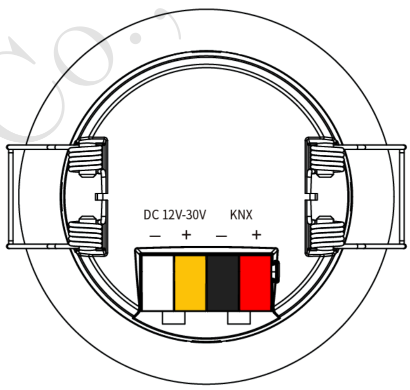

图 1.6: EDV21C-K 接口分布图
Fig1.6: EDV21C-K Interface Distribution Diagram

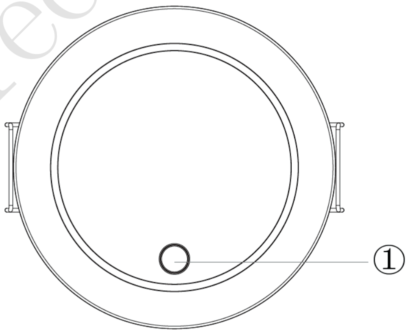

图 1.7: 按键分布图
Fig1.7: Key Distribution Map

表 1.1: EDV21C-W 接口功能表
Tab1.1: Interface Function Table

接口序号 Number	名称 Name	功能 Function
1	L	火线端口 Live wire port
2	N	零线端口 Zero-line port
3	N	零线端口 Zero-line port
4	L	火线端口 Live wire port

表 1.2: EDV21C-4 接口功能表
Tab1.2: Interface Function Table

接口序号 Number	名称 Name	功能 Function
1	V	DC 12-30V 电源输入正极 DC 12-30V power input positive terminal
2	G	电源负极 / GND Power negative terminal / GND
3	A	RS485-A
4	B	RS485-B

表 1.3: EDV21C-K 接口功能表
Tab1.3: Interface Function Table

接口序号 Number	名称 Name	功能 Function
1	KNX+(红) KNX+ (red)	KNX +
2	KNX-（黑） KNX- (Black)	KNX -
3	V（黄） V (Yellow)	12-30V DC 辅助供电 + 12-30V DC auxiliary power supply +
4	G（白） G (White)	12-30V DC -

表 1.4: 按键功能表
Tab1.4: Button Function Table

接口序号 Number	名称 Name	功能 Function
①	Set/Reset （-K：编程按键 Prog. key）	配网/复位共用按钮（-K 版为 KNX 编程按键） Network configuration / Reset (KNX programming key on -K)

1.5 性能参数 / Parameters & Performances

1.5.1 参数列表 / Parameters

表 1.5: 参数表
Tab1.5: Parameter Table

类别 Category	参数 Number	可配置？ Config.?	数值 Value	单位 Unit	说明 Remarks
雷达参数 RF Parameters	工作频段 Operating Freq.	N	59-64	GHz	毫米波频段 mmWave band
	发射功率 Tx Power	N	＜ 15	dBm	低功率毫米波发射 Low-power TX
	调制方式 Modulation	N	FMCW	-	调频连续波体制 FMCW scheme
	3dB 波束角 3dB Beamwidth	N	60°（xz 平面） 60°（yz 平面）	°	xz/yz 平面波束宽度 Beamwidth in xz/yz
	距离分辨率 Range Resolution	N	5	cm	距离维目标分辨能力 Range-axis resolution
性能参数 Specifications	感应范围 Sensing Range	Y	6×6 Max. （半径 3m）	m	挂高 3m 实测，范围可配 Measured at 3m, configurable
	可探测人数 Max. Persons	N	≤20	人	同时输出目标轨迹 With trajectory output
	运动灵敏度 Motion Sensitivity	Y	0-10 （默认 9）	级	小程序滑动条设置 Mini-program slider
	存在灵敏度 Presence Sensitivity	Y	0-10 （默认 9）	级	小程序滑动条设置 Mini-program slider
	雷达延时 Radar Delay	Y	0-1800 （默认 5）	s	检测保持延时 Detection hold
	目标消失延时 Target-miss Delay	Y	2-7200 （默认 5）	s	与雷达延时叠加生效 Stacked with radar delay
	上报周期 Reporting Period	Y	100-65500 （50 步进， 默认 1000）	ms	数据上报频率 Data reporting rate
	检测范围（X/Y） Range (X/Y)	Y	-350～+350 （默认）	cm	最小/最大值可配 min/max configurable
	检测高度（Z） Height (Z)	Y	0～350 （默认 30-200）	cm	最小/最大值可配 min/max configurable
电气参数 Electrical	工作电压 Operating Voltage	N	W：AC 90-270V 4：DC 12-30V K：DC 12-30V	V	分型号供电 Model-dependent
	工作电流 Current	N	＜ 800	mA	典型值 Typical
	待机功耗 Standby Power	N	2.64	W	典型值 Typical
物理参数 Physical Parameters	外形尺寸 Dimensions	N	Φ67×40	mm	开孔 Φ55，公差 ±0.2 Hole Φ55, tol. ±0.2
	重量 Weight	N	-	g	以实物为准 Per actual unit

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表 1.5: 参数表 (续)
Tab1.5: Parameter Table(continued)

类别 Category	参数 Number	可配置？ Config.?	数值 Value	单位 Unit	说明 Remarks
	外壳材质 Material	N	PC	-	-
	防护等级 IP Rating	N	IP67	-	室内防潮环境 Indoor moisture-proof
	安装方式 Mounting	N	嵌入式暗装 Embedded	-	开孔 Φ55，卡扣固定 Φ55 cutout, clips
	安装高度 Install Height	Y	0～4 （默认 3）	m	按实际挂高设置 推荐 2.5-3 m Set to actual, rec. 2.5-3 m
环境适应性 Environmental Adaptability	工作温度 Operating Temp.	N	-20～+60	℃	-
	储存温度 Storage Temp.	N	-20～+85	℃	-
	抗干扰类型 Anti-interference	N	光照、温湿度、 2.4G 无线设备等	-	不受光照温湿度干扰 Immune to light/temp/humidity
	抗震动 Vibration	N	-	-	远离震动源安装 Away from vibration

* 注：“可配置？”指的是是否提供接口供用户根据需要对该参数在允许范围内进行配置。
***Note:** ”Configurable?” means whether an interface is provided for users to set the parameter within the permissible range as required.

1.5.2 性能曲线 / Performances

* 注：以下性能曲线在室内空旷环境实测：安装高度 3m；X/Y 检测范围 -300～+300cm，检测高度 50-250cm（默认配置）；测试人员身高 170cm、体重 65-75kg，行走速度 1m/s。不同场景安装可能造成范围变化，以实际测试为准。
***Note:** The performance curves below were measured in an indoor open environment: installation height 3 m; X/Y detection range -300 to +300 cm and detection height 50 to 250 cm (default configuration); tester height 170 cm, weight 65-75 kg, walking speed 1 m/s. The range may vary in different installation scenarios, and actual on-site tests prevail.

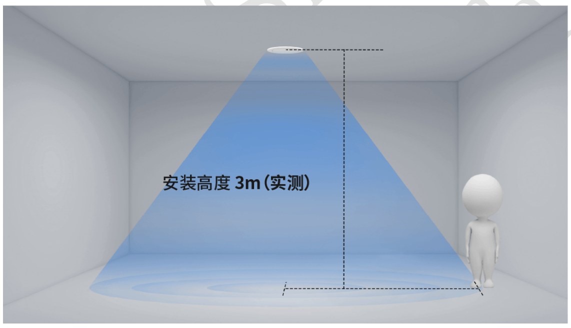

图 1.8: 挂高角度
Fig1.8: High-angle suspension

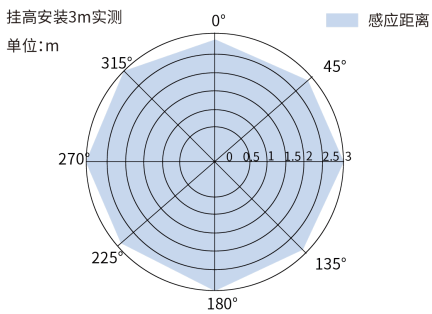

图 1.9: 感应性能
Fig1.9: Reaction performance

1.6 可定制项 / Specifications

表 1.6: 可定制项列表
Tab1.6: Customizable Items

定制项 **Items**	定制范围 **Customizable Range**	单位 **Unit**	说明 **Remarks**
安装高度 Mounting Height	0～400 （默认 300）	cm	与安装环境一致 Match actual install height
检测范围（X/Y） Detection Range (X/Y)	-350～+350 （默认）	cm	感应区坐标边界可自定义 Custom sensing-zone boundary
检测高度（Z） Detection Height (Z)	0～350 （默认 30-200）	cm	立体检测高度范围 Volumetric detection height
运动灵敏度 Motion Sensitivity	0-10 （默认 9）	级	数值越高越灵敏 Higher level, higher sensitivity
存在灵敏度 Presence Sensitivity	0-10 （默认 9）	级	数值越高越灵敏 Higher level, higher sensitivity
雷达延时 Radar Delay	0-1800 （默认 5）	s	检测保持时间 Detection hold time
目标消失延时 Target-missing Delay	2-7200 （默认 5）	s	人离开后保持时间 Hold time after departure
上报周期 Reporting Period	100-65500 （50 步进，默认 1000）	ms	数据上报频率 Data reporting rate
感应区 / 屏蔽区 Sensing / Shield Zones	各最多 10 个 三维长方体区域	-	适配现场检测边界与干扰屏蔽 Adapt boundaries and mask interference
通讯参数 Comm. Parameters	Wi-Fi、MQTT、透传等	-	按项目网络环境配置 Configure per project network

* 注：如有未标明的其他定制需求，请联系销售经理或 FAE 进行评估。
***Note:** Please contact your sales manager or FAE for evaluation if you have other unmarked customization requirements.

第二部分 / Chapter 2

命名规范 / Naming Rules

表 2.1: 命名规范示意
Tab2.1: Naming Nomenclature

制造商 Manufacture	频段 Frequency	大类别 Parent	子类别 Child	编号 Part No.	扩展功能 Extended Functions		流水号 Serial No.
ED	V	2	1	C		K	W/4
易探 EasyDetek	**C** 5.8GHz	**1** 雷达模组 RD Module	**0** 超低功耗 low power consumption	0 ～ 9 A ～ Z	**Y** 有光敏 With Photosensitive	**A** Matter	**1** Cat-1
	X 10.5GHz	**2** 独立传感器 Independent Sensor	**1** 旗舰 flagship		**N** 无光敏 Without Photosensitive	**C** Casambi	**2** 2.4G
	Q 24GHz	**3** 天线 & 器件 Antennas & Devices	**2** 侧装 Side-mounted		**G** 干接点 Dry node	**D** Dali	**4** RS485
	V 60GHz	**4** MCU	**3** 外置可调 External Adjustable		**S** AC 开关 AC Switch	**K** KNX	**B** BLE
	S 生态产品 Eco-products	**5** 配件 Accessories	**5** 通用 General			**M** 米家 Mi	**P** PLC
		6 IC	**6** 户外 Outdoor			**T** 涂鸦 Tuya	**R** RF433
		7 AI 传感 AI Sensor	**7** 保留 Reserved				**W** WiFi
		8 组网 Networking	**8** 基础 Basic				**Z** Zigbee
			9 高空 High Altitude				

示例： EDV21C-W = 易探（ED）+ 60GHz 频段（V）+ 独立传感器（2）+ 旗舰系列（1）+ 产品编号（C）+ WiFi 功能（W）；EDV21C-K 中 K 代表 KNX 功能，EDV21C-4 中 4 代表 485 功能。

Example: EDV21C-W = EasyDetek (ED) + 60GHz band (V) + Independent Sensor (2) + Flagship Series (1) + Part No. (C) + Wi-Fi (W); ”K” in EDV21C-K stands for KNX, and ”4” in EDV21C-4 stands for RS485.

第三部分 / Chapter 3
使用指南 / User Guide

3.1 安装指南 / Installation Guide

3.1.1 安装前检查 / Pre-installation inspection

- 确认安装环境符合要求，避免周边有震动源（冰箱、空调外机、水泵等）。
  Confirm that the installation environment meets the requirements: avoid any vibration sources around (refrigerators, air conditioner outdoor units, water pumps, etc.).
- 检查感应范围内无晃动物体、大面积金属物品（吊灯、风扇、金属涂层镜子等）。
  Check that there are no moving objects or large metal items within the sensing range (chandeliers, fans, metal-coated mirrors, etc.).
- 确认供电电源纹波低于 100mV，避免电源干扰。
  Confirm that the power supply ripple is below 100 mV to avoid power interference.
- 如与其他雷达传感器同场使用，确保安装间距大于 2 米。
  If used in the same area as other radar sensors, ensure the installation distance is more than 2 meters.

3.1.2 安装步骤 / installation procedure

- 安装高度：推荐 2.5-3 米，最大探测范围 4 米。
  Installation height: Recommended 2.5 - 3 meters, maximum detection height 4 meters.
- 嵌入式暗装：开设 Φ55mm 标准安装孔，提前布设供电 / 通讯线缆。
  Embedded concealed installation: Open a standard installation hole with a diameter of 55mm, and pre-arrange power / communication cables.

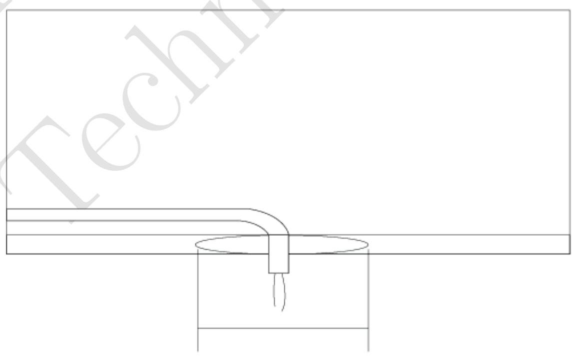

图 3.1: 天花板开孔 Φ55mm
Fig3.1: Ceiling hole Φ55mm

- 接线操作：按照对应型号接口定义，连接电源线、通讯线，确认接线牢固。
  Connection operation: According to the interface definitions of the corresponding model, connect the power cord and communication cable, and ensure that the connections are secure.

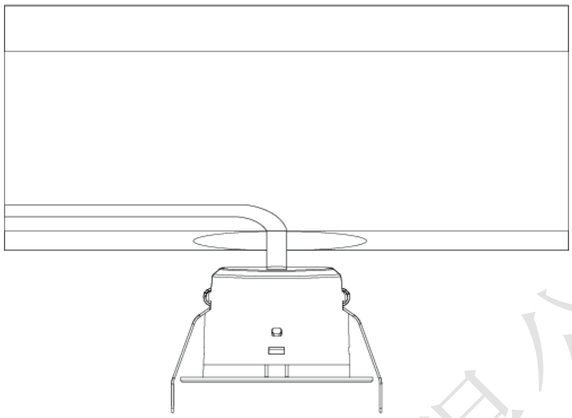

图 3.2: 设备与电源线连接
Fig3.2: Connect the equipment to the power cord.

- 卡扣调节：调整设备两侧安装卡扣，适配吊顶厚度。
  Adjustable clips: Adjust the installation clips on both sides of the equipment to fit the thickness of the ceiling.

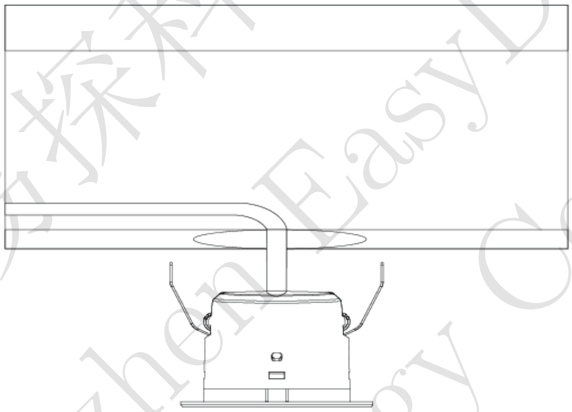

图 3.3: 调整安装卡扣
Fig3.3: Adjust the installation snap-on clips.

- 固定就位：将设备嵌入开孔，保持设备稳固，避免抖动，锁紧卡扣，完成安装。
  Fix and position: Insert the equipment into the opening, keep it stable, avoid shaking, lock the clips, and complete the installation.

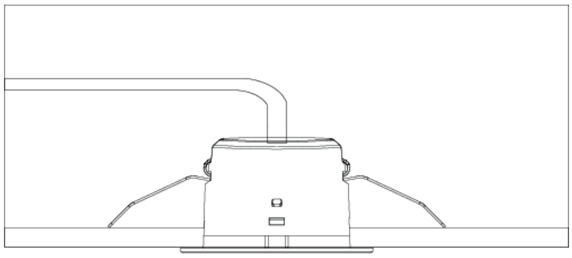

图 3.4: 嵌入固定，安装完成
Fig3.4: Embedded firmly, installation completed.

3.1.3 接线说明 / Connection Instructions

- EDV21C-W（Wi-Fi 版）：火线（L-L）与零线（N-N）两个端子互为连通状态，供电时选单火单零接入即可，如采用手拉手方式布线，可将另外两根线接入另外的 LN 端子，给其他传感器供电，不允许从感应器引线给高功率设备供电。
  EDV21C-W (Wi-Fi version): The two terminals, L-L (hot wire) and N-N (neutral wire), are in a connected state with each other. When powering, simply connect one hot wire and one neutral wire. If you use a hand-in-hand wiring method, you can connect the other two wires to another LN terminal to power other sensors. It is not allowed to supply power to high-power equipment from the sensor leads.

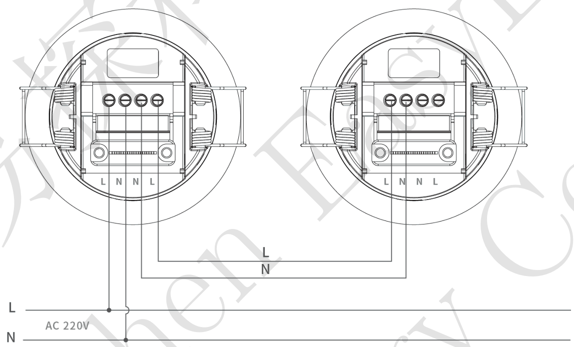

图 3.5: 供电范围：AC 220V 供电
Fig3.5: Power supply range: AC 220V supply

- EDV21C-4（RS485 版）：V（电源正）/ G（电源负）端子接 DC 12-30V 供电，注意极性；A/B 端子接 RS485 总线，多机 Modbus 组网时按手拉手（菊花链）方式连接总线。
  EDV21C-4 (RS485 version): Connect the V (positive) and G (negative) terminals to a DC 12-30V power supply, observing the polarity; connect the A/B terminals to the RS485 bus. For multi-device Modbus networking, wire the bus in a hand-in-hand (daisy-chain) topology.

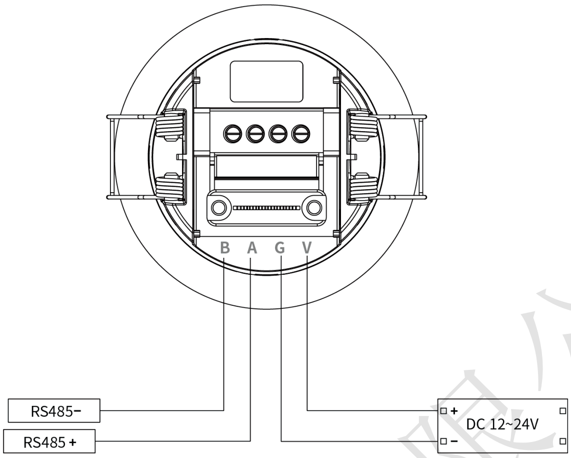

图 3.6: 供电范围：DC 12-30V 供电，RS485 通信接口（A/B 端子）
Fig3.6: Power supply range: DC 12-30V power supply, RS485 communication interface (ports A/B)

- EDV21C-K（KNX 版）：KNX+（红）/ KNX-（黑）接 KNX 总线（按线色区分极性）；V（黄）/ G（白）端子接 DC 12-30V 辅助供电；① 键为 KNX 编程按键，供 KNX 调试工具使用。
  EDV21C-K (KNX version): Connect KNX+ (red) / KNX- (black) to the KNX bus (polarity by wire color); connect V (yellow) / G (white) to a DC 12-30V auxiliary power supply. The ① key is the KNX programming key for KNX commissioning tools.

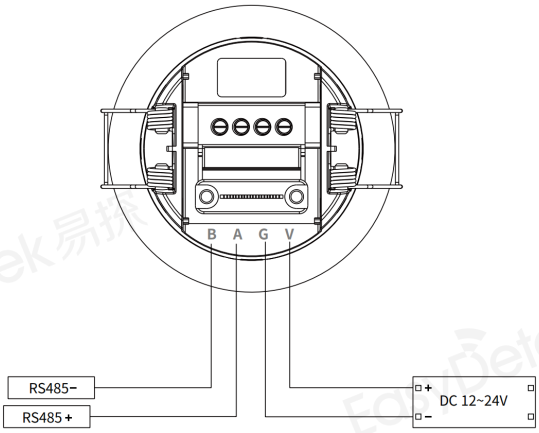

图 3.7: 供电范围：DC 12-30V 供电，对接 KNX 系统专用通讯线路
Fig3.7: DC 12-30V power supply, connect the dedicated communication lines for the KNX system

3.2 功能时序 / timing sequence

表 3.1: 全系列统一指示灯与功能时序
Tab3.1: Unified indicator lights and function timing for the entire series

序号 Number	工作状态 Working condition	指示灯说明 Indicator Light Explanation
1	上电初始化 Power-on initialization	通电后绿色指示灯常亮 5 秒 After powering on, the green indicator light remains on for 5 seconds
2	感应触发 Inductive trigger	识别到人体移动 / 存在， 绿色指示灯每 5 秒闪烁 1 次 Detects human movement / presence, and the green indicator light flashes once every 5 seconds
3	MQTT 连接成功 MQTT connected	红色指示灯闪烁 1 次（带 Wi-Fi 功能版本） The red indicator light flashes once (for versions with Wi-Fi)
4	配网模式/恢复出厂设置 Distribution network mode / Reset to factory settings	长按配网键 5 秒后，1 秒闪 4 次进入配网模式， 配网过程中每 3 秒闪烁 1 次 After holding down the network distribution key for 5 seconds, it flashes 4 times per second to enter the network distribution mode. During the network distribution process, it flashes once every 3 seconds
5	延时关闭 Delayed closure	目标离开后，按设定的目标消失延时结束后恢复待机 After the target leaves, it resumes standby mode after the set target disappearance delay ends

3.3 设置和调试指南 / Debugging and Setting

3.3.1 初始化与自学习 / Initialization and Self-learning

设备首次安装、更换安装环境、修改分区参数后，建议断电重启一次，设备自动完成环境自学习，过滤墙体、固定家具等静态干扰，保障人数统计、轨迹定位准确性；新增 / 修改感应区、屏蔽区后，保存配置等待 10 秒算法生效，再现场测试探测效果。

After the equipment is first installed, the installation environment is changed, and the partition parameters are modified, it is recommended to power off and restart the device once. The device will automatically complete the environment self-learning process, filtering out static interference such as walls and fixed furniture, ensuring the accuracy of personnel counting and trajectory positioning. After adding or modifying the sensing area and shielding area, save the configuration and wait for 10 seconds for the algorithm to take effect, then conduct on-site testing of the detection effect.

3.4 小程序使用指南 / Mini Program User Guide

3.4.1 连接准备 / Connection Preparation

- 设备正常上电，开启手机蓝牙与定位，搜索对应设备蓝牙名称完成连接。
  The device is powered on normally. Turn on the Bluetooth and location of the mobile phone, search for the corresponding Bluetooth name of the device to complete the connection.
- 进入 EDV21C 设备控制主界面，查看设备基础信息。
  Enter the main interface of the EDV21C device control to view the basic information of the device.

3.4.2 主界面功能 / Main interface functions

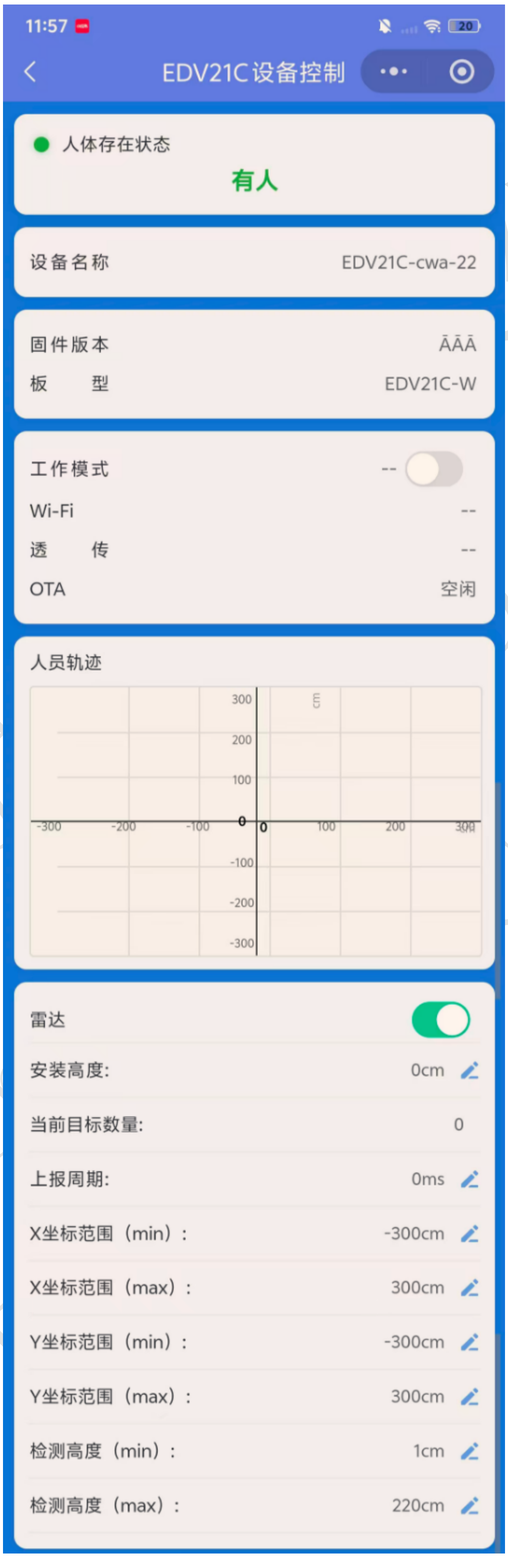

图 3.8: 蓝牙小程序功能说明
Fig3.8: Function Description of Bluetooth Mini Program

人体存在状态显示 / Human Presence Status Display：小程序首页显示人体存在状态，能直观区分“有人/无人”，用于快速判断当前检测区域内是否存在人员。
The mini-program home page displays the human presence status, clearly distinguishing ”occupied/vacant” for quick judgment of whether anyone is present in the current detection area.

主要功能 / Main Functions：

- 设备信息显示：设备名称、固件版本、板型等基础信息，便于售后与调试确认。
  Device information: device name, firmware version, board type, etc., for after-sales and debugging confirmation.
- 工作及连接状态显示：工作模式、Wi-Fi 连接状态、透传状态、OTA 状态等运行信息。
  Operating and connection status: working mode, Wi-Fi connection, passthrough and OTA status.
- 人员轨迹显示：以坐标图形式显示目标轨迹及点位、当前目标数量，可查看目标 X/Y 坐标。
  Trajectory display: target trajectories and positions on a coordinate map, target count, and X/Y coordinates.
- 雷达开关控制：在小程序中开启或关闭雷达检测功能。
  Radar on/off control: enable or disable radar detection in the mini-program.
- 安装高度设置：按实际安装高度设置（单位 cm，可设 0 400，默认 300）。
  Installation height setting: set per actual height (in cm, range 0-400, default 300).
- 上报周期设置：坐标、目标数量及状态数据的上报频率（100-65500ms，50 步进，默认 1000ms）。
  Reporting period setting: reporting rate of coordinates, target count and status data (100-65500 ms in 50 ms steps, default 1000 ms).
- 检测范围设置：X/Y 坐标范围最小/最大值（-350～+350）、检测高度最小/最大值（可设 0 350，默认 30-200）。
  Detection range setting: X/Y min/max values (-350 to +350) and min/max detection height (up to 350, default 30-200).
- 感应区 / 屏蔽区管理：新增、从设备同步区域，并显示当前区域数量（各最多 10 个）。
  Sensing / shield zone management: add and sync zones, display zone count (up to 10 zones each).
- 运动灵敏度 / 存在灵敏度设置：以滑动条形式设置（0-10 级，默认 9）。
  Motion / presence sensitivity setting: set via slider (0-10 levels).
- 雷达延时 / 目标消失延时设置。
  Radar delay / target-missing delay setting.
- Wi-Fi 配置（SSID、密码）；MQTT 配置（主机、端口、周期、开关）。
  Wi-Fi configuration (SSID, password); MQTT configuration (host, port, period, switch).
- 透传调试：透传 Host / 端口配置，写入透传并切换模式、切回正常模式。
  Passthrough debugging: host/port configuration, enter passthrough mode and switch back.
- 配置保存 / 导入 / 同步 / 断开，便于批量调试与配置复用。
  Configuration save / import / sync / disconnect, for batch debugging and reuse.

3.4.3 注意事项 / Notice

- Wi-Fi 密码仅写入设备，页面不回显，谨防密码泄露。
  Wi-Fi passwords are only written to the device and the page does not echo. Please be cautious of password leakage.

- 修改通讯、分区等核心参数后，建议同步保存配置，避免断电丢失。
  After modifying core parameters such as communication and partitioning, it is recommended to save the configuration synchronously to avoid loss in case of power failure.
- 蓝牙连接距离建议控制在 5 米以内，障碍物过多会导致连接不稳定。
  It is recommended that the Bluetooth connection distance be controlled within 5 meters. Excessive obstacles may cause unstable connection.

第四部分 / Chapter 4
协议规范 / Protocol Specification

4.1 通信接口 / Communication Interface

表 4.1: 接口列表
Tab4.1: Interface List

接口类型 Interface Type	EDV21C-W	EDV21C-4	EDV21C-K	功能说明 Functional Specification
蓝牙 BLE	√	√	√	小程序调试、参数配置、OTA 升级 Mini-program debugging, parameter setting and OTA upgrade
UART 串口	√	√	√	本地调试、产测、数据透传 Local debugging, production test, and data passthrough
RS485 总线	×	√	×	Modbus RTU，工业 / 楼宇总线组网 Modbus RTU, industrial/building bus networking
KNX 总线	×	×	√	KNX 楼宇总线专用接口 Dedicated KNX building-bus interface
Wi-Fi	√	√	√	MQTT 云上报、OTA、远程透传 MQTT cloud reporting, OTA, and remote passthrough

*** 注：** EDV21C 全系保留 Wi-Fi/MQTT 传输能力（-W 为基础联网方式），-4 / -K 型号分别以 RS485 / KNX 总线为主要通讯方式。
***Note:** The whole EDV21C series retains Wi-Fi/MQTT capability (the primary connection for -W), while the -4 / -K models mainly use the RS485 / KNX bus respectively.

4.2 通讯协议 / Communication Protocol

- 蓝牙协议：私有蓝牙透传协议，专供小程序配对、参数配置、实时数据查看。
  Bluetooth Protocol: Private Bluetooth transparent transmission protocol, specifically designed for mini-program pairing, parameter setting, and real-time data viewing.
- UART 串口协议：标准串口通讯，支持本地指令读写、参数查询、产测与原始数据输出。
  UART Serial Port Protocol: Standard serial port communication, supporting local instruction reading and writing, parameter query, production testing, and raw data output.
- RS485 协议：Modbus RTU 总线协议，支持多设备组网、轮询读写参数与检测数据。
  RS485 Protocol: Modbus RTU bus protocol, supporting multi-device networking, polling for reading and writing parameters, and detection data.

- KNX 协议：兼容标准 KNX 楼宇通讯协议，适配主流 KNX 智能家居、楼宇自动化系统。
  KNX Protocol: Compatible with the standard KNX building communication protocol, it is adapted to mainstream KNX smart home and building automation systems.

- MQTT 协议：基于 Wi-Fi 实现，支持设备状态、人数统计、目标坐标与轨迹数据云端上报，上报周期可自定义。
  MQTT Protocol: Implemented based on Wi-Fi, it supports cloud-based reporting of device status, people counting, target coordinates and trajectory data, with a customizable reporting period.

*** 注**：详细协议指令、寄存器地址、数据帧格式，请参考配套独立协议文档：《EDV21C MQTT 数据传输协议文档》《EDV2XX Modbus RTU 协议说明》《EDV21C-K KNX 数据传输协议文档》《EDV21C 蓝牙小程序开发接口文档》。

***Note:** For detailed protocol instructions, register addresses, and data frame formats, please refer to the accompanying standalone protocol documents: EDV21C MQTT Data Transmission Protocol, EDV2XX Modbus RTU Protocol Description, EDV21C-K KNX Data Transmission Protocol, and EDV21C BLE Mini-Program Interface Document.

第五部分 / Chapter 5
售后服务 / Aftersale service

5.1 安全合规 / Safety Compliance

- 本产品符合室内智能传感设备安规标准，仅适用于室内环境使用。
  This product complies with the safety standards for indoor intelligent sensing devices and is only suitable for use in indoor environments.
- 电气接线严格遵循国标电气规范，非专业人员禁止拆机、改线。
  Electrical wiring strictly follows the national standard electrical specifications. Non-professionals are prohibited from disassembling the machine or modifying the wiring.
- 设备防护等级 IP67，仅适用于常规室内防潮环境，禁止长期浸泡、户外淋雨使用。
  The equipment has a protection grade of IP67 and is only suitable for regular indoor moisture-proof environments. Long-term immersion and outdoor rain exposure are strictly prohibited.

5.2 免责声明 / Disclaimers

- 因用户私自拆机、改装接线、违规供电造成的设备损坏、安全事故，本公司不承担保修及相关责任。
  Our company will not be held responsible for the warranty and related liabilities of equipment damage and safety accidents caused by users' private disassembly of the machine, modification of wiring, or illegal power supply.
- 未按照安装规范（间距、远离干扰源、避开金属区域）安装，导致设备误触发、检测异常，不在免费维保范围内。
  Failure to install the equipment in accordance with the installation specifications (spacing, distance from interference sources, and avoidance of metal areas), resulting in false triggering or abnormal detection of the equipment, is not covered by the free maintenance.
- 自然灾害、人为磕碰、外力损坏、进水进尘造成的故障，不予免费保修。
  Faults caused by natural disasters, human bumps, external force damage, water or dust ingress are not covered by free warranty.
- 产品固件、规格会持续迭代优化，更新版本不再单独通知，如需最新资料请联系销售人员。
  The product firmware and specifications will be continuously iterated and optimized. Updated versions will not be notified separately. If you need the latest information, please contact the salesperson.
- 测试参数为实验室标准环境数据，实际使用效果因现场环境、安装方式存在差异，不属于产品质量问题。
  The test parameters are standard environmental data from the laboratory. The actual usage effect may vary due to differences in on-site environment and installation methods, and does not fall under product quality issues.

5.3 售后服务 / After-Sales Service

- 质保服务：产品按照行业标准提供质保，质保期内非人为质量问题可享受维修、更换服务。
  Warranty service: The product is covered by warranty in accordance with industry standards. During the warranty period, non-human-caused quality issues can be repaired or replaced.
- 技术支持：安装调试、参数配置、协议对接问题，可联系我司销售或 FAE 工程师提供技术支持。
  Technical support: For installation and commissioning, parameter configuration, and protocol integration issues, please contact our sales or FAE engineers for technical support.
- 定制服务：个性化功能、参数、接口定制，可对接商务团队评估方案与工期。
  Customized services: Personalized functions, parameters, and interface customization, which can be integrated with business teams to evaluate plans and project schedules.
- 资料获取：最新版说明书、协议文档、固件程序，请联系官方工作人员索取。
  For information acquisition: The latest version of the instruction manual, protocol documents, and firmware programs, please contact the official staff for them.

附录 A 系列清单 / Series Family

表 A.1: EDV21C 系列型号清单
Tab1.1: EDV21C Series Model List

型号 **Model**	供电 **Power Supply**	主要通讯 **Primary Communication**	备注 **Remarks**
EDV21C-W	AC 90-270V	Wi-Fi（MQTT） 蓝牙 BLE	基础款 Base model
EDV21C-4	DC 12-30V	RS485（Modbus RTU） 蓝牙 BLE	保留 Wi-Fi/MQTT 能力 Retains Wi-Fi/MQTT
EDV21C-K	DC 12-30V	KNX 总线 蓝牙 BLE	KNX 编程按键；保留 Wi-Fi/MQTT 能力 KNX programming key; retains Wi-Fi/MQTT

*** 注：**全系统一 59-64GHz 毫米波探测、人数统计与轨迹追踪（$\leq$20 人）、蓝牙小程序调试、嵌入式暗装（安装开孔 Φ55mm）。

***Note:** The whole series shares 59-64GHz mmWave detection, people counting and trajectory tracking (up to 20 targets), Bluetooth mini-program debugging, and embedded ceiling installation (Φ55mm cutout).

附录 B 应用案例 / Application Cases

B.1 案例 1：智慧办公 / Case 1: Smart Office

开放式办公区按工位分区部署 EDV21C，实时统计各分区在场人数并识别人员有无，联动照明与空调：有人自动开启、全员离开后按设定延时关闭，实现按需用能管理。三维分区屏蔽功能可排除工位隔断、柜体等静态干扰，保证统计准确。Deploy EDV21C units zone by zone in open office areas to count occupants and detect presence in real time, linking lighting and air conditioning: they switch on automatically when occupied and switch off after the set delay once everyone leaves, enabling on-demand energy management. The 3D zone shielding function filters static interference such as workstation partitions and cabinets to ensure accurate counting.

B.2 案例 2：智慧酒店 / Case 2: Smart Hotel

客房顶部嵌入安装 EDV21C，识别入住有人/无人状态与室内人数，联动客房用电节能管控与精准服务提醒。设备无摄像头、无图像采集，从根源保护住客隐私，满足酒店行业对隐私合规的严格要求。EDV21C is embedded in the guest-room ceiling to identify occupancy status and the number of people indoors, linking with in-room energy-saving control and precise service reminders. With no camera and no image capture, guest privacy is fundamentally protected, meeting the strict privacy-compliance requirements of the hotel industry.